| CONDITIONS | RULES | | | |
|---|---|---|---|---|
| Is the date administered < absolute minimum age date? | Yes | No | No | No |
| Is the absolute minimum age date ≤ date administered < minimum age date? | No | Yes | No | No |
| Is the minimum age date ≤ date administered < maximum age date? | No | No | Yes | No |
| Is the date administered ≥ maximum age date? | No | No | No | Yes |
| | | | | |
| **OUTCOMES** | No. The vaccine dose administered was not administered at a valid age for the target dose. Evaluation reason is 'Too young'. | Yes. The vaccine dose administered was administered at a valid age for the target dose. Evaluation reason is 'Grace period'. | Yes. The vaccine dose administered was administered at a valid age for the target dose. | No. The vaccine dose administered was not administered at a valid age for the target dose. It is extraneous. Evaluation reason is 'Too old'. |

**TABLE 6-16 EVALUATE AGE BUSINESS RULES**

| Business Rule ID | Business Rule |
|---|---|
| CALCDTAGE-1 | A patient's maximum age date must be calculated as the patient's date of birth plus the maximum age. |
| CALCDTAGE-4 | A patient's minimum age date must be calculated as the patient's date of birth plus the minimum age. |
| CALCDTAGE-5 | A patient's absolute minimum age date must be calculated as the patient's date of birth plus the absolute minimum age. |

## 6.5 Evaluate Preferable Interval

*Evaluate preferable interval* validates the date administered of a vaccine dose administered against defined preferable interval(s) from previous vaccine dose(s) administered or other events. In cases where a target dose does not specify preferable interval attributes, the interval is considered “valid.”

Preferable intervals can be measures in four different ways:

- “From Immediate Previous Dose Administered Flag” requires the interval to be evaluated from the immediate previous vaccine dose administered and is used in the majority of cases.
- “From Target Dose # in Series” requires the interval to be evaluated from the date of the specified dose.
- “From Most Recent Vaccine Type” requires the interval to be evaluated from the date of the most recently administered dose of any of the specific vaccine types listed (e.g., this is used in Pneumococcal to ensure proper spacing between the different intervals between PCV13 and PPSV23).

- “From Relevant Observation Code” requires the interval to be evaluated from the date of a particular patient observation (e.g. the interval for a dose of Pertussis vaccine is measured from the date of the onset of pregnancy).

It is possible for a given dose to use multiple preferable interval types. For example, dose 3 of HepB and dose 3 of HPV, each have two preferable intervals. The first interval is from the immediate previous vaccine dose administered. The second interval is from satisfied target dose 1 in each respective antigen series. Note that if multiple intervals are specified, then all intervals must be satisfied in order for the dose to satisfy the interval requirements.

Figure 6-7 provides the evaluation interval timeline used to define adjacent intervals by using *from immediate previous dose administered flag* as the reference point.

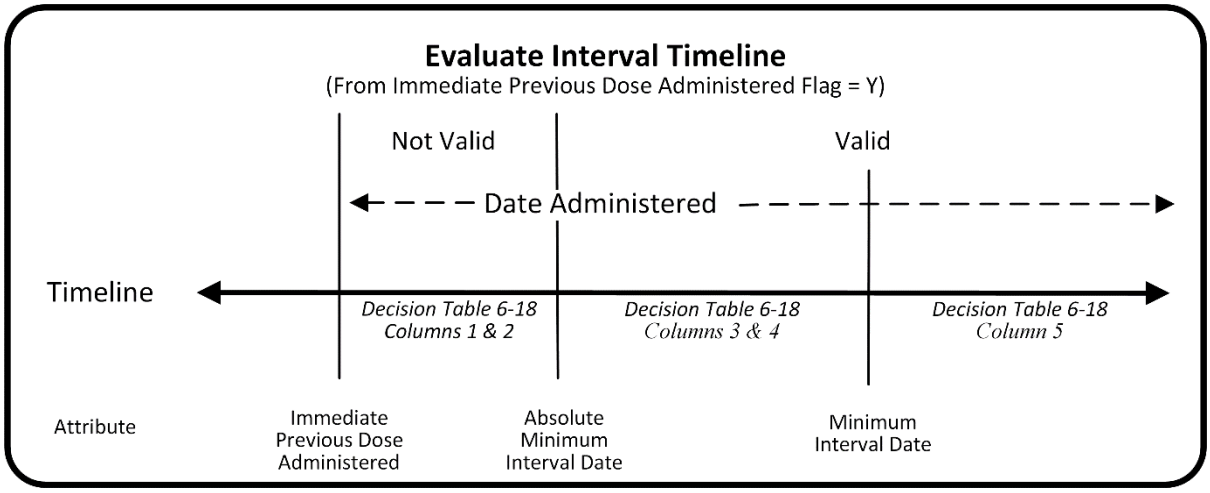


**FIGURE 6-7 EVALUATE INTERVAL ‘FROM IMMEDIATE PREVIOUS DOSE ADMINISTERED FLAG’ TIMELINE**

Figure 6-8 illustrates the evaluation interval timeline used to define non-adjacent intervals by using *from target dose number in series* as the reference point. This timeline is used only when from immediate previous dose administered flag is “N”.

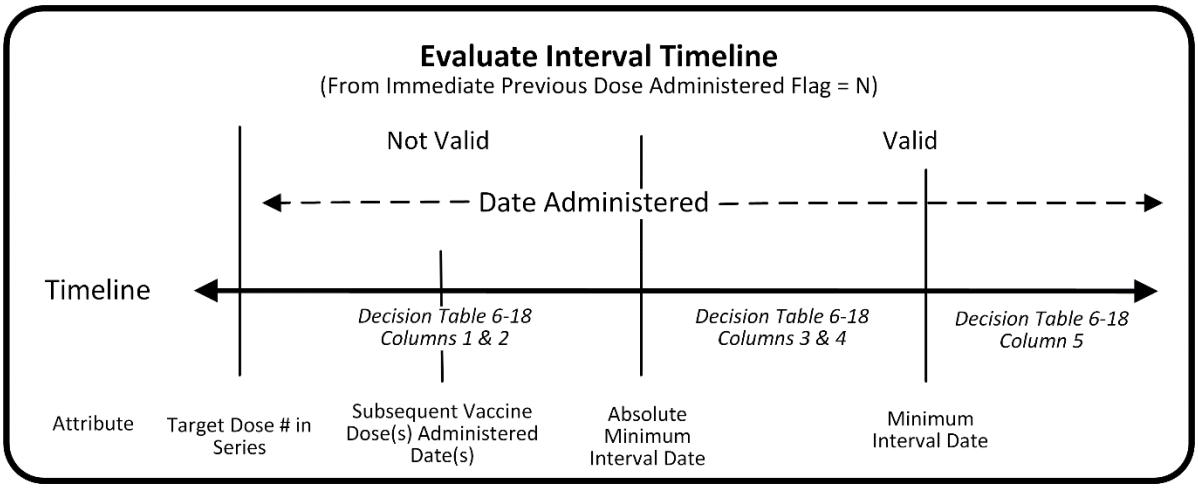


**FIGURE 6-8 EVALUATE INTERVAL 'FROM TARGET DOSE NUMBER IN SERIES' TIMELINE**

Figure 6-9 illustrates the evaluation interval timeline used to define most recent vaccine intervals by using *from most recent vaccine type* as the reference point.

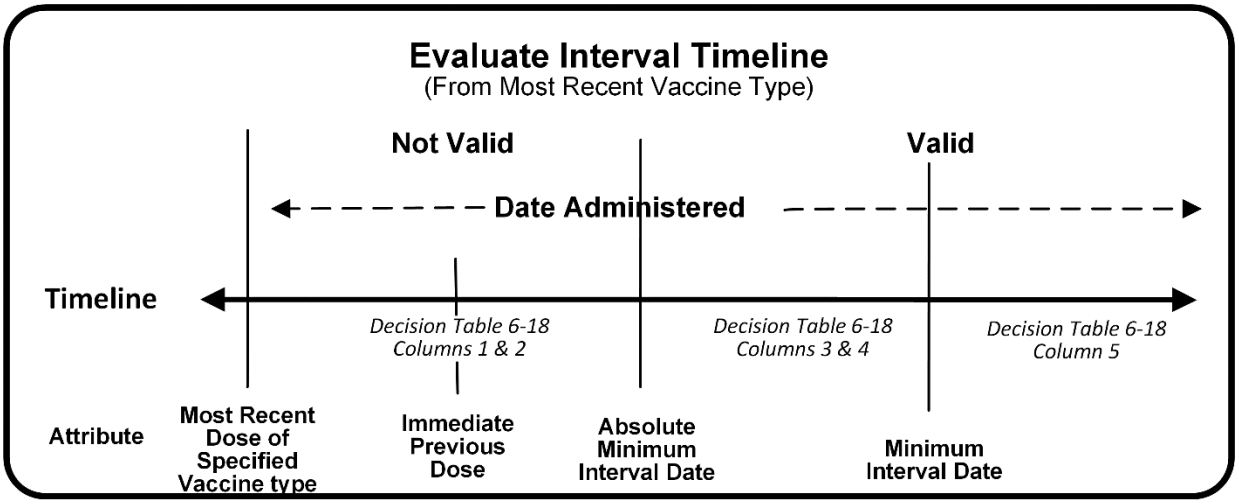


**FIGURE 6-9 EVALUATE INTERVAL 'FROM MOST RECENT VACCINE TYPE' TIMELINE**

Figure 6-10 illustrates the evaluation interval timeline used to define most recent vaccine intervals by using *from relevant observation code* as the reference point.

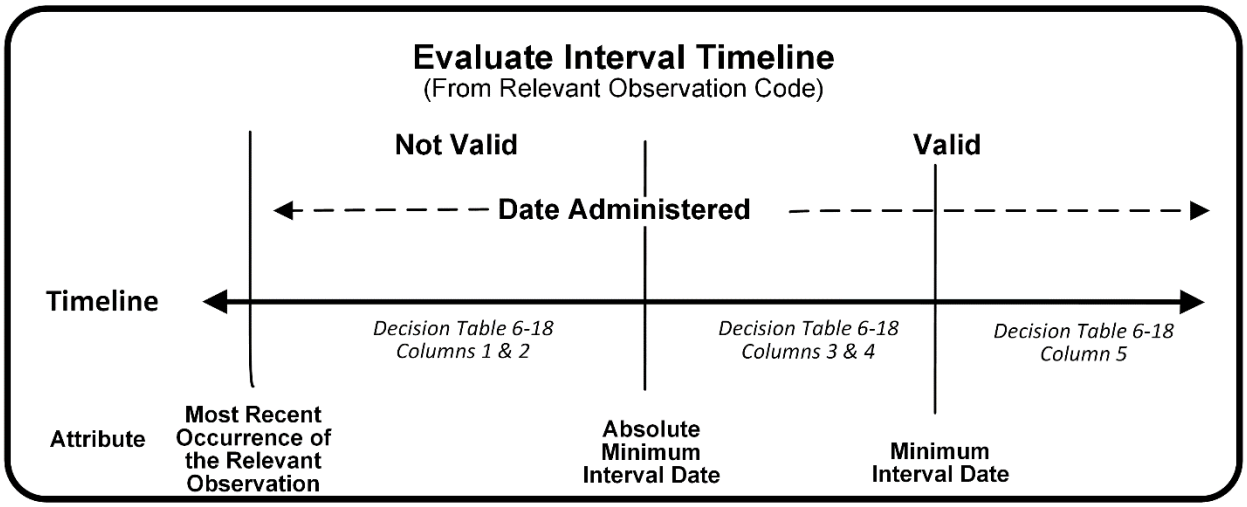


**FIGURE 6-10 EVALUATE INTERVAL 'FROM RELEVANT OBSERVATION CODE' TIMELINE**

The following process model, attribute table, decision table, and business rule table are used to evaluate preferable interval of a vaccine dose administered.

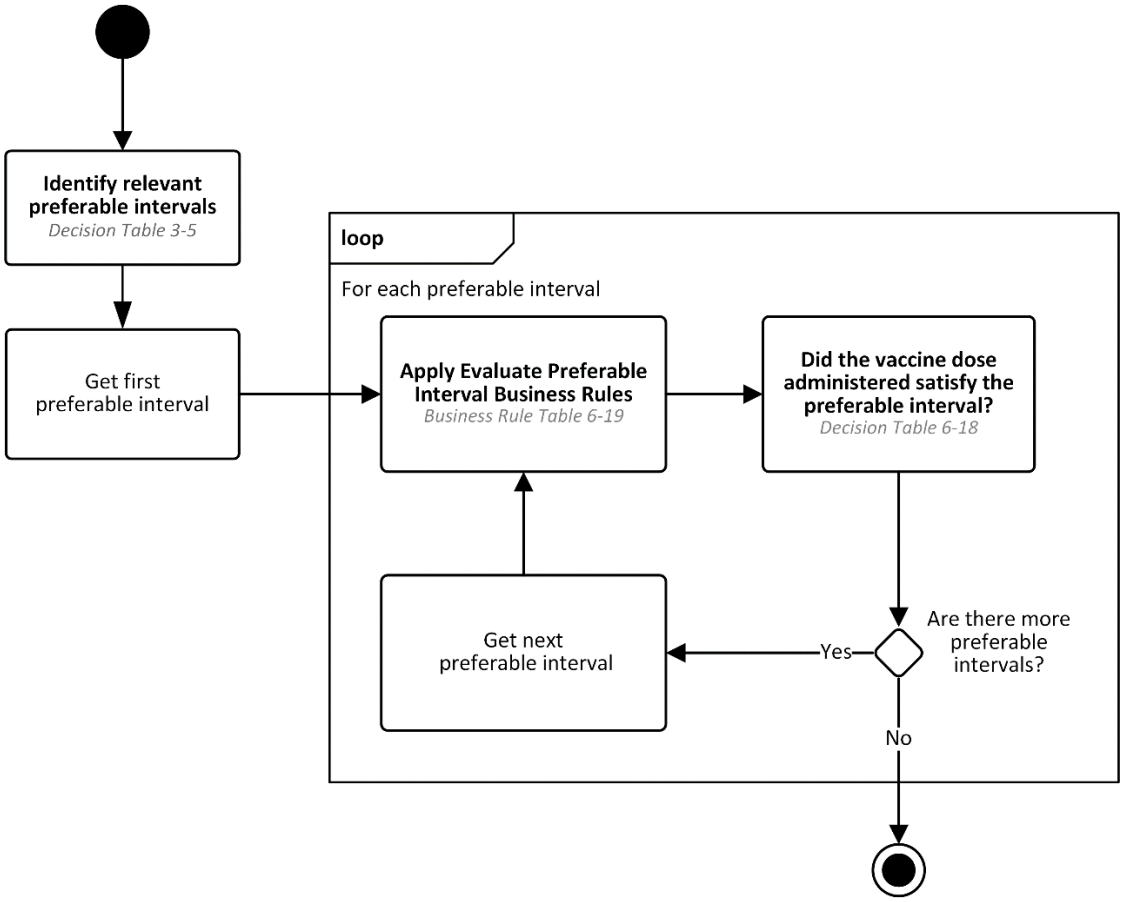


**FIGURE 6-11 EVALUATE PREFERABLE INTERVAL PROCESS MODEL**

**TABLE 6-17 PREFERABLE INTERVAL ATTRIBUTES**

| Attribute Type | Attribute Name | Assumed Value if Empty |
|---|---|---|
| Vaccine dose administered | Date Administered | - |
| Supporting Data | Preferable Interval elements | - |
| Calculated date (CALCDTINT-3) | Absolute Minimum Interval Date | 01/01/1900 |
| Calculated date (CALCDTINT-4) | Minimum Interval Date | 01/01/1900 |

**TABLE 6-18 DID THE VACCINE DOSE ADMINISTERED SATISFY THE PREFERABLE INTERVAL FOR THE TARGET DOSE?**

| CONDITIONS | RULES | | |
|---|---|---|---|
| Is the date administered < absolute minimum interval date? | Yes | No | No |
| Is the absolute minimum interval date ≤ date administered < minimum interval date? | No | Yes | No |

| CONDITIONS | RULES | | |
|---|---|---|---|
| Is the minimum interval date ≤ date administered? | No | No | Yes |
| | | | |
| **OUTCOMES** | No. The vaccine dose administered did not satisfy the preferable interval for the target dose. Evaluation reason is 'Too soon'. | Yes. The vaccine dose administered satisfied the preferable interval for the target dose. Evaluation reason is 'Grace period'. | Yes. The vaccine dose administered satisfied the preferable interval for the target dose. |

**Table 6-19 Evaluate Preferable Interval Business Rules**

| Business Rule ID | Business Rule |
|---|---|
| CALCDTINT-1 | A patient's reference dose date for an interval must be calculated as the date administered of the most immediate previous vaccine dose administered if all the following are true:<br>• the interval has a from immediate previous dose administered flag of 'Y'<br>• the vaccine dose administered has an evaluation status of 'Valid' or 'Not Valid'<br>• the vaccine dose administered is not an inadvertent administration. |
| CALCDTINT-2 | A patient's reference dose date for an interval must be calculated as the date administered of the vaccine dose administered that satisfies the target dose with the same target dose number as the from target dose number in series if all the following are true for the interval:<br>• from immediate previous dose administered flag is 'N'<br>• from target dose number in series is not 'N/A'. |
| CALCDTINT-3 | A patient's absolute minimum interval date must be calculated as the patient's reference dose date plus the absolute minimum interval. |
| CALCDTINT-4 | A patient's minimum interval date must be calculated as the patient's reference dose date plus the minimum interval. |
| CALCDTINT-8 | A patient's reference dose date for an interval must be calculated as the date administered of the most recent vaccine dose administered that is the same vaccine type as the from most recent vaccine type if all the following are true for the interval:<br>• from immediate previous dose administered flag is 'N'<br>• from most recent vaccine type is not 'N/A'<br>• the vaccine dose administered is not an inadvertent administration. |
| CALCDTINT-9 | A patient's reference dose date for an interval must be calculated as the observation date of the most recent active patient observation if all the following are true for the interval:<br>• from immediate previous dose administered flag is 'N'<br>• from relevant observation code is not 'N/A'. |

## 6.6 Evaluate Allowable Interval

*Evaluate allowable interval* validates the date administered of a vaccine dose administered against defined allowable interval(s) from previous vaccine dose(s) administered. In rare cases, intervals can be applied which are either abnormally early – usually specified in ACIP footnotes or subsequent clarifications – or intervals which differ following a not valid administration.